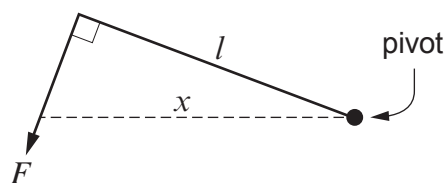


2. (a) The following question has been set in a previous Physics exam:

Explain what is meant by the moment of a force about a pivot.

Five of the responses given are shown below (A to E). **Only three of the responses are correct.** Place a tick (✓) in the boxes next to the correct responses. [2]

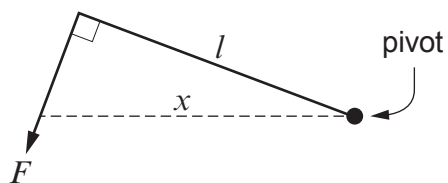
A.



$$\text{moment} = Fl$$

☐

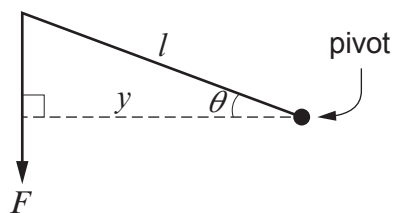
B.



$$\text{moment} = Fx$$

☐

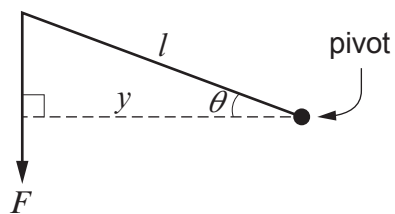
C.



$$\text{moment} = Fl \cos \theta$$

☐

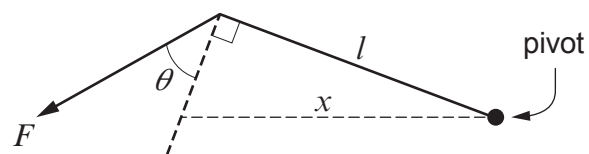
D.



$$\text{moment} = Fl$$

☐

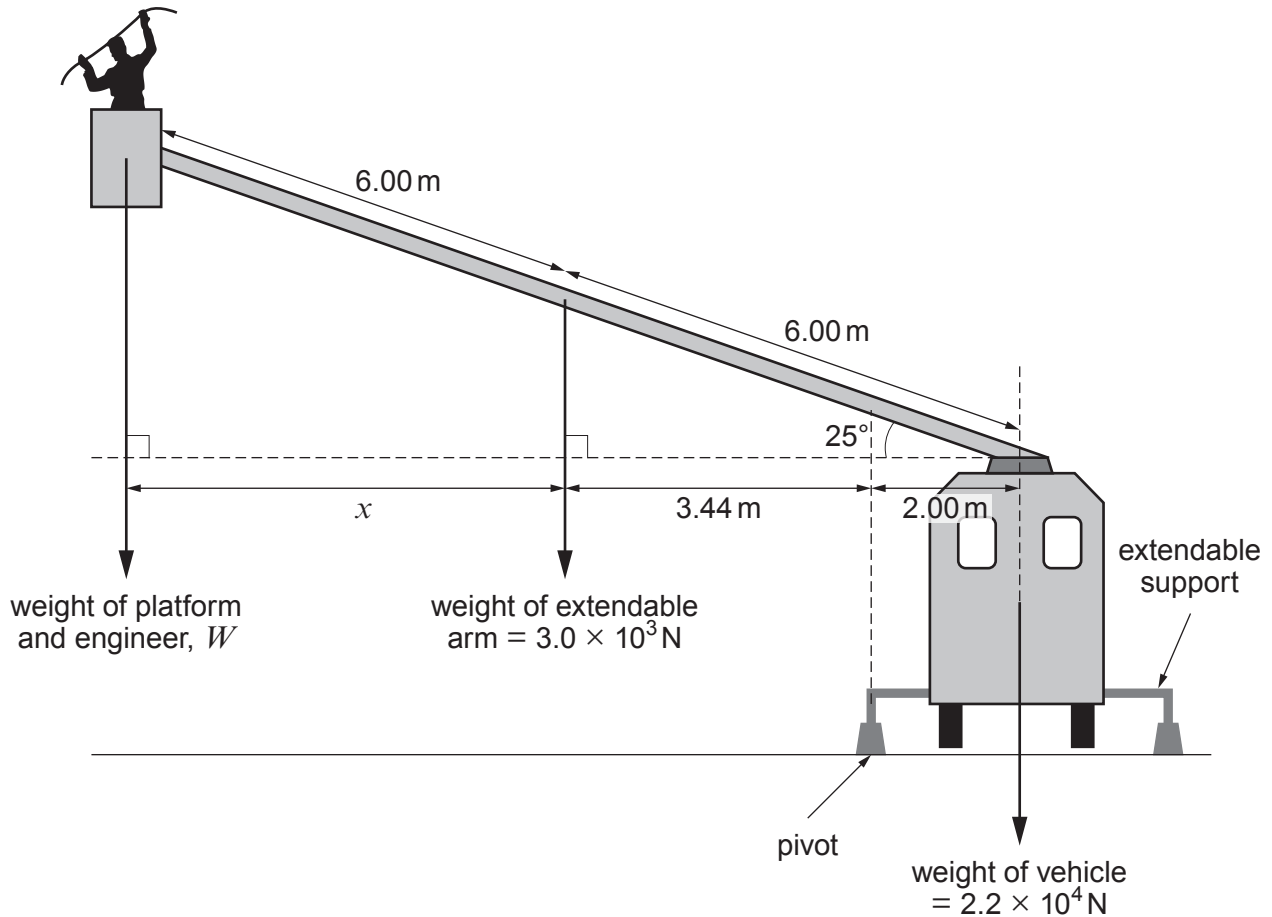
E.



$$\text{moment} = F \cos \theta l$$

☐


- (b) Telephone engineers sometimes use an extendable arm to repair telephone lines. The arm is able to extend from the top of a specially modified vehicle. When the arm is in use extendable supports are used as shown. The pivot point is also shown.



- (i) Show that the distance, x , in the diagram is approximately 5.4 m. [1]

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- (ii) Determine the maximum possible weight of the platform and engineer, W , in the situation shown in the diagram, so that the system does not topple. [3]

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Examiner
only

(iii) A sign placed on the platform gives the following warning:

Maximum 2 persons or 200 kg

Consider whether or not this is a reasonable warning to give. Assume the arm is at its maximum extension in the diagram and the angle shown (25°) is the lowest possible. The weight of the platform is 1700 N. [4]

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